



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2024

Paper: Electronics-II

Course Code: PHYS-3702

Roll No. 05-10

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- i. Convert $(001010100100)_2$ to hexadecimal.
- ii. Write 279 in excess 3 code. 0101010100
- iii. Is NAND gate associative or not?
- iv. Differentiate between amplifier and oscillator.
- v. Convert (10011100) gray code to binary. 11101000
- vi. Convert $(354)_8$ to hexadecimal.
- vii. Why does the Class C amplifier prefer over other amplifiers?
- viii. Convert the given expression into standard form.
 $X = (A+C)(A+B)$
- ix. Differentiate between multivibrator and flip-flop.
- x. Write down the expression for power gain of common collector amplifier in terms of ratio of resistances.
- xi. What is a bandwidth of an amplifier when upper critical frequency = 5kHz and lower critical frequency = 250Hz?
- xii. List the amplifier stages in a typical operational amplifier.
- xiii. What are the applications of operational amplifier?
- xiv. At what point is a class C amplifier normally biased?
- xv. What is push-pull operation in class B amplifier?

Q.2. Answer the following questions.

(3x10=30)

- I. Analyze the bipolar junction transistor amplifier for total low frequency response.
 - a. Determine the lower critical frequency and phase shift of input RC circuit.
 - b. Determine the lower critical frequency and phase shift of output RC circuit.
 - c. Determine the lower critical frequency of bypass RC circuit.
 - d. Describe a Bode plot.
- II. Briefly discuss the ways to connect negative feedback with operational amplifier. What are the benefits of negative feedback in operational amplifiers?
- III. What is oscillator? Discuss working principle of oscillator. Name the two types of oscillators.

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A B C D E



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Fall 2023

Roll No. 045516

Paper: Electronics-II
Course Code: PHYS-3702

(Common with BS-AD 6th)

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

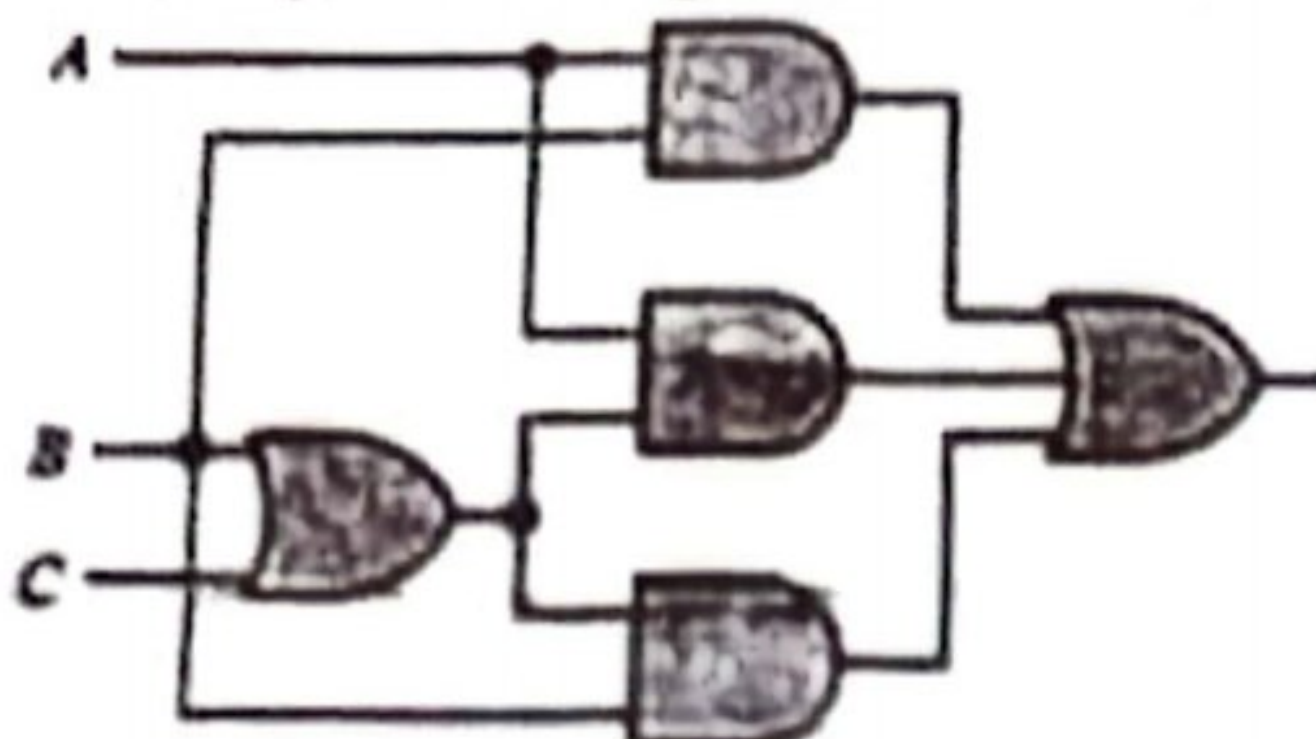
Q.1. Answer the following short questions.

(15x2=30)

- i. What is multivibrator?
- ii. Define Excess-3 code with example.
- iii. Convert $(00111010)_2$ to hexadecimal.
- iv. Convert $(798)_{10}$ to binary.
- v. Is NOR gate associative or not?
- vi. Convert $(011010111101)_2$ into gray code.
- vii. Differentiate between amplifier and oscillator.
- viii. Name the two types of oscillators.
- ix. How does the Class AB amplifier differ from a Class B amplifier?
- x. Convert the given expression into standard form.
 $X = AB + ABD + BCD$
- xi. What are two types of clipping in class A amplifier?
- xii. Define bandwidth.
- xiii. What is miller effect?
- xiv. Differentiate inverting and non-inverting configuration of operational amplifier.
- xv. What are the conditions required to start oscillation in circuit?

Q.2. Answer the following questions.

- I. Analyze the bipolar junction transistor amplifier for total high frequency response.
 - (a) Determine the upper critical frequency and phase shift of input RC circuit.
 - (b) Determine the upper critical frequency and phase shift of output RC circuit.
 - (c) Describe a Bode plot. (10)
- II. Briefly discuss the effect of negative feedback on input and output impedance of different configuration of operational amplifier. What are the benefits of negative feedback in operational amplifiers. (10)
- III. (a) Verify the following expression using Boolean algebra.
 - a. $A + A.B = A$
 - b. $(A+B).(A+C) = A + B.C$
 - c. $A + \bar{A}.B = A + B$
 (b) find the minimized sum of product expression for the given circuit diagram using Boolean algebra.



(5+5)

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THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (6x5=30)

- (i) Explain why the gain of RC coupled amplifier reduces at low and high frequencies?
- (ii) What is the effect if the emitter resistor R_E is not bypassed in a C-E amplifier? Give details.
- (iii) In Hartley oscillator $L_1 = 250 \mu\text{H}$, $L_2 = 50 \mu\text{H}$ and $C = 300 \text{ pF}$. Find the value of its frequency.
- (iv) Define class B condition in power amplifiers. Where would you establish the Q-point on the characteristic curves?
- (v) Subtract the following hexadecimal numbers:
 a. $(C3)_{16} - (0B)_{16}$ b. $(84)_{16} - (2A)_{16}$
- (vi) By using laws of Boolean algebra, prove that $A + \bar{A}B = A + B$

Answer the following questions.

Q.2. (a) Discuss the low frequency response of RC coupled common emitter (C-E) amplifier by determining expression for voltage gain $A_{V(\text{low})}$.

(b) Explain how negative feedback stabilizes the gain in an amplifier. (6, 4)

Q.3. (a) Draw the circuit and explain the operation of Monostable multivibrator. Find the relation for pulse width in it.

(b) Draw the circuit and discuss the analysis of Operational Amplifier as non inverting amplifier. (6, 4)

Q.4. (a) Simplify the expression by using Boolean algebra.

$$\overline{AB + AC} + \bar{A}\bar{B}C$$

(b) Map the following SOP expressions on a Karnaugh map:

$$A\bar{B}CD + \bar{A}B\bar{C}\bar{D} + AB\bar{C}D + ABCD + AB\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}C\bar{D}$$

(6, 4)



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- i. ✓ Convert $(001010100100)_2$ to hexadecimal.
- ii. Write 279 in excess 3 code.
- iii. Is NAND gate associative or not?
- iv. ✓ Differentiate between amplifier and oscillator.
- v. ✓ Convert (10011100) gray code to binary.
- vi. ✓ Convert $(354)_8$ to hexadecimal.
- vii. ✓ Why does the Class C amplifier prefer over other amplifiers?
- viii. ✓ Convert the given expression into standard form.
$$X = (A+C)(A+B)$$
- ix. ✓ Differentiate between multivibrator and flip-flop.
- x. Write down the expression for power gain of common collector amplifier in terms of ratio of resistances.
- xi. ✓ What is a bandwidth of an amplifier when upper critical frequency = 5kHz and lower critical frequency = 250Hz?
- xii. ✓ List the amplifier stages in a typical operational amplifier.
- xiii. ✓ What are the applications of operational amplifier?
- xiv. ✓ At what point is a class C amplifier normally biased?
- xv. ✓ What is push-pull operation in class B amplifier?

Q.2. Answer the following questions.

(3x10=30)

- I. Analyze the bipolar junction transistor amplifier for total low frequency response.
 - a. Determine the lower critical frequency and phase shift of input RC circuit.
 - b. Determine the lower critical frequency and phase shift of output RC circuit.
 - c. Determine the lower critical frequency of bypass RC circuit.
 - d. Describe a Bode plot.
- II. Briefly discuss the ways to connect negative feedback with operational amplifier. What are the benefits of negative feedback in operational amplifiers?
- III. What is oscillator? Discuss working principle of oscillator. Name the two types of oscillators.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions:

(6x5=30)

- (i) What is the effect if the emitter resistor R_E is not bypassed in a C-E amplifier? Give details.
- (ii) What is an oscillator? Describe two requirements that must be satisfied to make an oscillator from a feedback amplifier?
- (iii) In an Astable multivibrator $R_1 = 2.7 \text{ K}\Omega$, $C_1 = 1000 \text{ pF}$, $R_2 = 5.2 \text{ K}\Omega$, $C_2 = 1500 \text{ pF}$. Find the value of its frequency.
- (iv) Define class A condition in power amplifiers. Where would you establish the Q-point on the characteristic curves?
- (v) Do the following conversions.
 - a. $(650)_{10} = (?)_{16}$
 - b. $(11000110)_2 = (?)_{\text{Gray}}$
- (vi) What are Universal gates? Use only NAND gates to implement AND gate.

Answer the following questions.

(5x6=30)

- Q.2. (a) What is Miller-effect? Draw the high frequency equivalent circuit for the RC coupled common emitter (C-E) amplifier and determine expression for Miller input capacitance C_{ie} .
- (b) Draw the C-E amplifier's frequency response curve and explain why its gain reduces at low and high frequencies? (6, 4)
- Q.3. (a) Define negative feedback. Drive the relation between gain with feedback and gain without feedback.
- (b) Draw the circuit and describe the working of Operational Amplifier when it is used as an integrator circuit. (6, 4)
- Q.4. (a) Use a Karnaugh map to reduce the following expression to a minimum SOP form:
$$\overline{A}\overline{B} + A\overline{B} + \overline{C}\overline{D} + C\overline{D}$$
- (b) Consider the following Boolean expression $Y = (AB)(\overline{B} + C)$, and construct a truth table for it. (6, 4)



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – 2021
Paper: Electronic Devices and Circuits
Course Code: PHY-304-A

Part – II

Roll No. 028512

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following:

(4x5=20)

- (i). Discuss briefly which diode exhibits negative resistance?
- (ii). Draw the circuit diagrams for the inverting and non-inverting configurations of an Operational Amplifier.
- (iii). What is Cross-over distortion in power amplifiers?
- (iv). What factors determine f_2 for the high-frequency range in a C-E amplifier?
- (v). What two requirements must be satisfied to make an oscillator from a feedback amplifier?

Answers the following questions.

(3x10=30)

Q.3. (a) Discuss the operation of a Capacitor Filter when it is connected to a full-wave center-tapped rectifier circuit. Derive an expression for its ripple factor.

(b) What is the effect of negative feedback on bandwidth of an amplifier?

(8, 2)

Q.4. (a) Discuss the low frequency response of a single stage RC coupled C-E amplifier. Draw its equivalent circuit and find out expressions for its voltage gain and phase angle.

(b) Define class A condition in power amplifiers. Where would you establish the Q-point on the characteristic curves?

(7, 3)

Q.5. (a) Draw the circuit of practical Hartley oscillator, explain its working and calculate the frequency of oscillation.

(b) Draw the circuit of an Astable multivibrator using NPN transistors and also sketch its output waveforms.

(6, 4)

$$38.7 = 46$$

$$12$$

$$8.7$$

$$3.7$$



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program: Fifth Semester – 2020

Paper: Electronic Devices and Circuits
Course Code: PHY-304-A

Part – II

Roll No. [REDACTED]
Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following: (4x5=20)

- i. Describe briefly why the gain of RC coupled amplifier reduces at low and high frequencies?
- ii. What is meant by Op-Amp? Draw the circuit diagram of a summing Op-Amp.
- iii. Describe oscillator feedback principles. What is the effect of negative feedback on bandwidth of an amplifier?
- iv. Discuss briefly the crystal control of frequency in a certain crystal oscillator.
- v. Differentiate between light emitting diode and a photo diode.

Answers the following questions.

(3x10=30)

Q.3: (a)

Sketch the high frequency equivalent circuit for the common emitter (C-E) amplifier and derive a relation for the Miller-effect capacitance C_{le} .

(b)

What function does the bypass capacitor serve in a C-E amplifier?

(8, 2)

Q.4: (a)

Determine the theoretical power conversion efficiency in class A power amplifier.

(b)

Define class B condition. Where would you establish the Q point?

(7, 3)

Q.5: (a)

Describe the circuit operation of a practical RC phase shift oscillator.

(b)

Draw the circuit and output waveforms for a Monostable multivibrator.

(6, 4)



UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No. 019599

PAPER: Electronic Devices and Circuits

Course Code: PHY-304-A Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2: Write short answers of the following questions: (5×4 = 20)

- Describe briefly why the gain of RC coupled amplifier reduces at low and high frequencies?
- What two requirements must be satisfied to make an oscillator from a feedback amplifier?
- What function does the C_C and C_E serve in a common emitter (C-E) BJT amplifier?
- Describe briefly under what conditions do we use a π filter and π -R filter?
- How does a Zener diode regulate dc voltage?

Q.3: (a) Draw the circuit and discuss the high frequency response of RC coupled BJT amplifier by determining $A_{V(high)}$.
(b) Find out an expression for the phase angle $\theta_{(high)}$ of above circuit. (8, 2)

Q.4: (a) Describe circuit operation of a class B power amplifier and find out an expression for its theoretical power conversion efficiency.
(b) Draw the circuit diagram of an Astable multivibrator. (8, 2)

Q.5: (a) Draw the circuit of a practical Colpitts oscillator and determine its frequency of oscillations.
(b) A transistor Colpitts oscillator has $C_1 = C_2 = C$ and $L = 150 \mu H$. Find $C_1 = C_2$ for oscillations at 1.5 M Hz. (7, 3)



UNIVERSITY OF THE PUNJAB

Fifth Semester 2018
Examination: B.S. 4 Years Programme

Roll No. 214865

PAPER: Electronic Devices and Circuits
Course Code: PHY-304-A

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE TYPE

Q.2 Short Answer

5*4= 20

1. What is β_{OC} and α_{OC} of the transistor if $I_C = 20.3\text{mA}$ and $I_E = 20.5\text{mA}$.
2. Explain the function of a Zener Diode, where we use it.
3. What is the Amplification? What are the factors to determine the Voltage gain of an Amplifier?
4. When a transistor is used as a switch, in what two states is it operated?
5. Draw the circuit of Common Collector Amplifier, how it work.

Q.3 Draw the circuit of Schmitt-trigger circuit, How it work where it used?

6

b) What is a Miller Effect?

4

Q.4 What is the meaning of negative feedback, what its importance in an Oscillator circuits? Write the function of a Phase shift Oscillator, what type of wave at the output?

10

Q.5 Draw the Class AB Amplifier, write down its function, where it used?

10



UNIVERSITY OF THE PUNJAB

Fifth Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Electronic Devices and Circuits
Course Code: PHY-304-A

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Note: Attempt all Questions

Q.2 Write short answers of the following questions:

(2*10=20)

- I. What is the difference between π -Filter and π -R Filter? Explain it with circuits.
- ✓ II. What is difference between Positive and Negative feed-back, which one is used for Oscillator?
- III. What is a Schmitt trigger circuit, what is its function?
- IV. What is a Miller effect?
- ✓ V. What is a Tunnel diode?
- VI. Discuss basic idea of Light Emitting Diode.
- ✓ VII. What is the meaning of Multivibrator, what are its types, which one is the pulse generator?
- VIII. Define and explain the Ripple factor.
- IX. What is a power transistor, where we used it?
- X. What is an OP-Amp? What are its characteristics?

✓ Q.3 a) What are basic components of an Oscillator circuit?

2

b) Draw and explain the function of Phase Shift Oscillator, what type of wave it generates? 8 ✓

Q.4 a) Explain why the coupling capacitors do not have a significant effect on gain at sufficiently high signal frequencies?

3

b) Explain the working of Bi-stable Multivibrator.

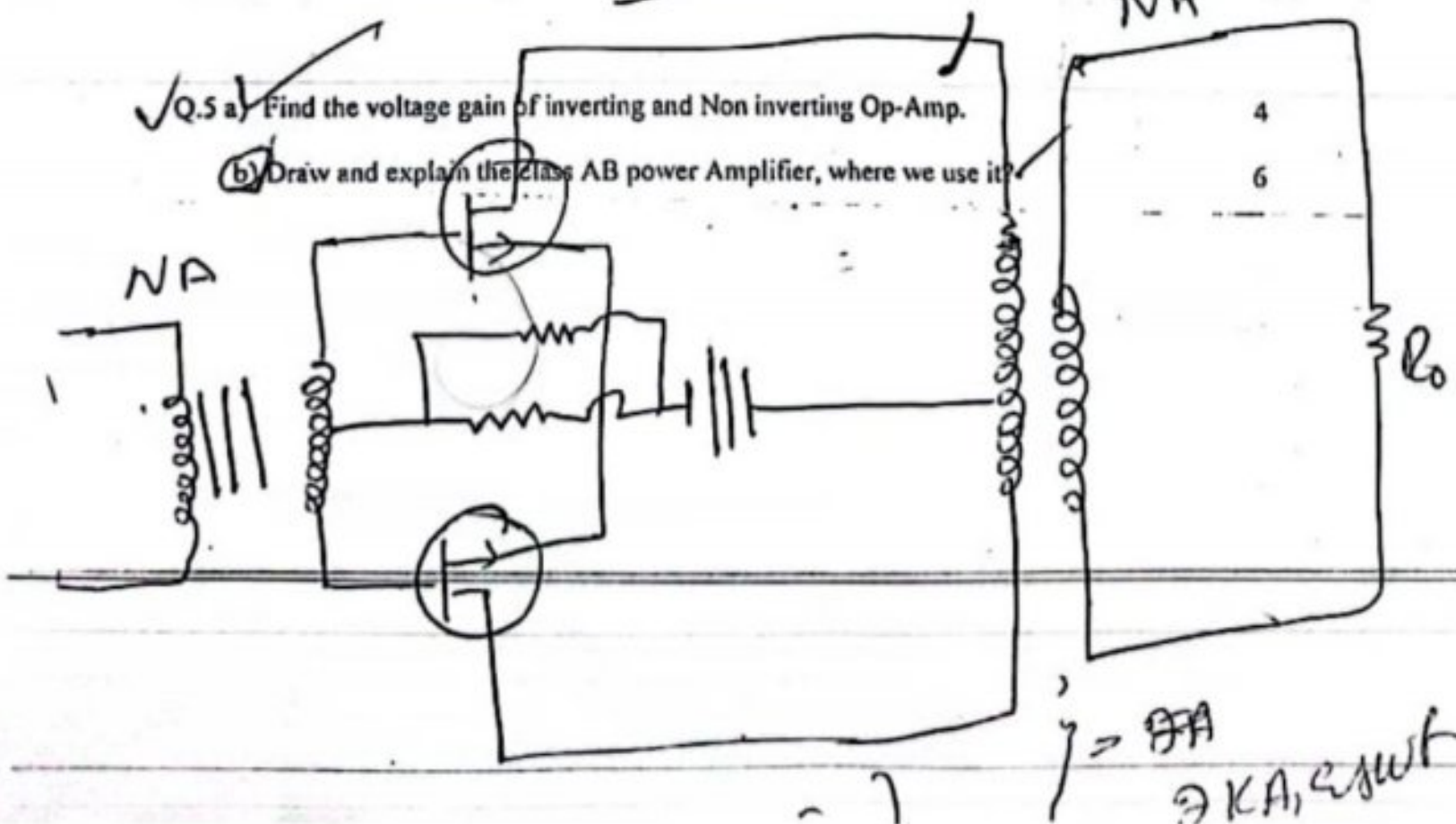
7

✓ Q.5 a) Find the voltage gain of inverting and Non inverting Op-Amp.

b) Draw and explain the Class AB power Amplifier, where we use it?

4

6





UNIVERSITY OF THE PUNJAB

Fifth Semester 2015
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Electronic Devices and Circuits
Course Code: PHY-304/304A

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Short Questions - Max Marks: 20

- Q.1 Write down the expression for Harmonic distortion as well as for total Harmonic distortion (%THD). *Ans: $P_{THD} = \frac{P_{2nd} + P_{3rd} + \dots}{P_{fund}}$* 3
- Q.2 Give a comprehensive idea of voltage regulation of a power supply. *Ans: $V_{reg} = \frac{V_{no-load} - V_{full-load}}{V_{full-load}} \times 100\%$* 3
- Q.3 Draw and label the frequency response of an amplifier. *Ans: $f_{L} = \frac{1}{2\pi R_i C_i}$, $f_{H} = \frac{1}{2\pi R_o C_o}$* 3
- Q.4 Describe I/V characteristics of a Tunnel diode comprehensively. *Ans: $I = I_0 \left(e^{-\frac{qV}{kT}} - 1 \right)$* 3
- Q.5 What does a ripple factor quantify? *Ans: $r = \frac{V_{ripple}}{V_{dc}}$* 2
- Q.6 For both inverting and non-inverting configuration of an operational amplifier calculate the close loop gain when $R_1 = 1k\Omega$ and $R_2 = R_f = 1M\Omega$. *Ans: $A_v = -\frac{R_f}{R_1}$ for inverting, $A_v = 1 + \frac{R_f}{R_1}$ for non-inverting* 2
- Q.7 What do you know of feedback ratio β ? *Ans: $\beta = \frac{V_o}{V_i}$* 2

Full Questions - Max Marks: 30

- Q.1 (a) What do you understand by Miller effect capacitance with regards to the frequency response of an amplifier. Evaluate its expression reflected at the input side of an amplifier. It is applicable to common emitter amplifiers only, explain. *Ans: $C_{in} = C_{gs} + C_{gd} \left(1 + |A_v| \right)$* 2+5
- (b) List down the fundamental characteristics of an Operational amplifier. *Ans: $A_{OL} \rightarrow \infty$, $f_{L} \rightarrow 0$, $f_{H} \rightarrow \infty$, $R_i \rightarrow \infty$, $R_o \rightarrow 0$* 3
- Q.2 Discuss the operation of class B amplifier and calculate its dc input power, ac output power, efficiency and also the power dissipated by the output transistors with maximum power consideration. *Ans: $P_{dc} = I_{C(sat)} V_{CEQ}$, $P_{ac} = \frac{1}{2} I_{C(sat)} V_{CEQ}$, $\eta = \frac{P_{ac}}{P_{dc}} = 78.5\%$, $P_{D1} = P_{D2} = \frac{1}{4} I_{C(sat)} V_{CEQ}$* 10
- Q.3 (a) List criteria necessary for a circuit to oscillate. *Ans: $\angle \phi = 0^\circ$ or 360° , $|A\beta| \geq 1$* 3
- (b) Discuss in detail the Wein Bridge Oscillator. Calculate its frequency of oscillations and gain at which this circuit will oscillate. *Ans: $f = \frac{1}{2\pi RC}$, $A = 3$* 7

$$P_{out} = V_{(rms)} I_{rms}$$

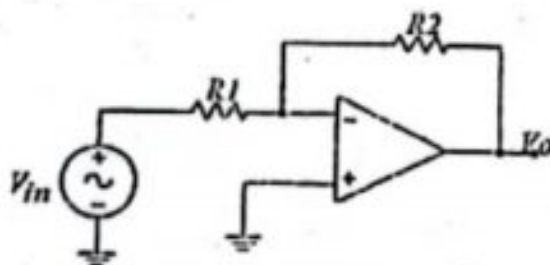
$$P = 0.25 \cdot I_{C(sat)} V_{CEQ}$$

$$V_{CC} = 2V_{CEQ}$$

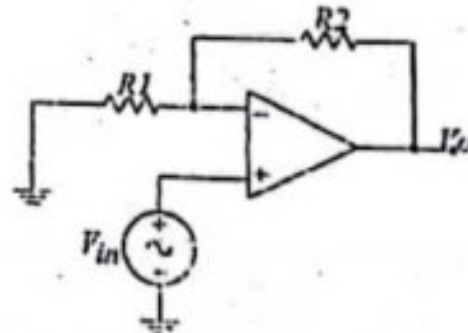
Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPEShort Questions Max Marks: 20

- Q.1 Differentiate between zener and Avalanche breakdown. 3
- Q.2 Give a comprehensive idea of voltage regulation of a power supply. 3
- Q.3 Calculate the efficiency of a class B amplifier for a supply voltage of $V_{CC} = 20\text{ V}$ with peak reverse voltage of $V_L(p) = 17\text{ V}$. 3
- Q.4 Calculate the percentage ripple of a capacitor filtered power supply, operated from a 60 Hz, 120-V_{rms} main line. It has a 100 μF filter capacitor and load $R_L = 1\text{-K}\Omega$. 3
- Q.5 Discuss the basic principle of a light emitting diode. 4
- Q.6 For the following circuits, what value of R_1 will be required if $R_2 = R_f = 100\text{k}\Omega$, $v_o = 10\text{ V}_{rms}$ and $v_{in} = 1\text{ V}_{rms}$. 4



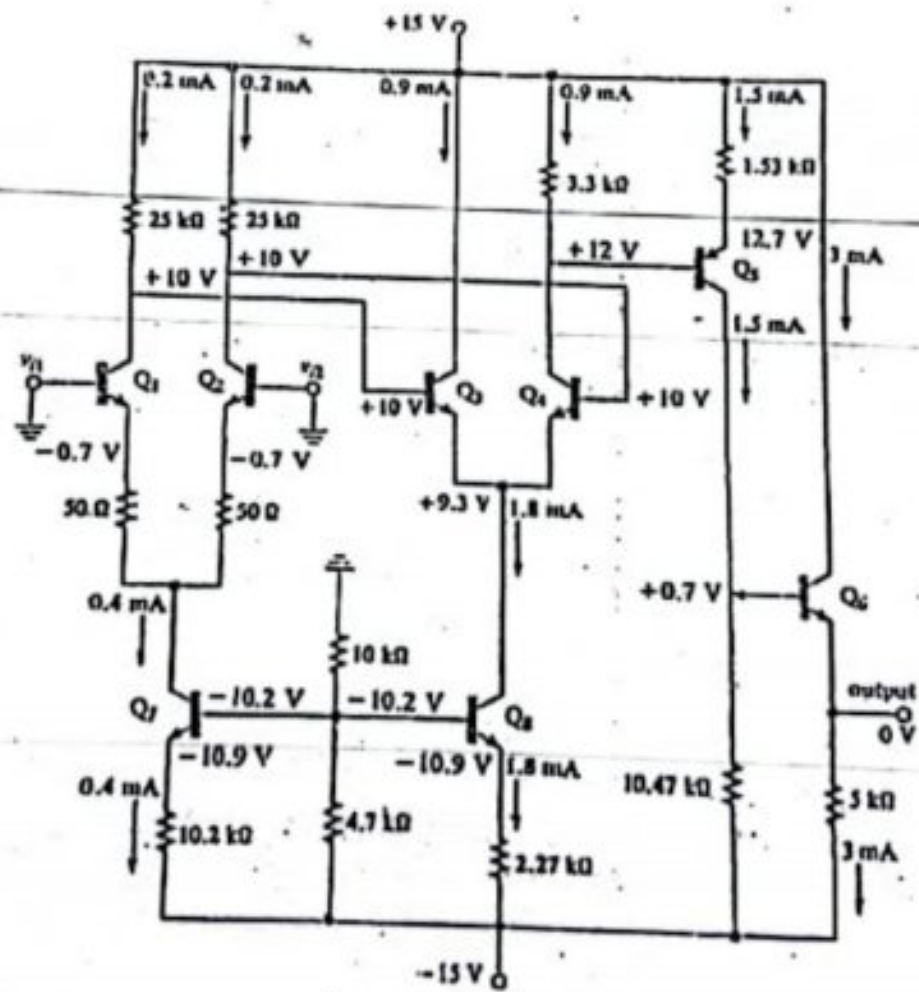
(a) Inverting amplifier



(b) Non-inverting amplifier

(P.T.O.)

- Q.1 (a) Describe very comprehensively how series capacitors result in gain reduction in low frequency region in BJT amplifier and show that the lower cut off frequency due to C_1 can be given by $f_L(C_1) = 1/2\pi(r_s + r_{in})C_1$. 3+4
- (b) Show that lower and upper cut off points in BJT amplifier i.e. f_L and f_H are also termed as half power points. 3
- Q.2 Describe very comprehensively the dc operation of an Operational Amplifier shown below? 10
- Q.3: (a) Describe very comprehensively the feedback concept in Non-inverting amplifier and evaluate its closed loop gain. 7
- (b) Show by diagram that feedback ratio is same i.e. $\beta = R_1 / (R_1 + R_2)$ for both inverting and non-inverting amplifiers. 3



PAPER: Electronic Devices and Circuits
Course Code: PHY-304TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Full Questions

Max Marks: 39

Q.1 (a) What is an Operational amplifier and write down its important characteristics.

Notes

(b) Describe the operation of a series fed class A amplifier and evaluate its output power and efficiency.

Q.2 (a) Evaluate the expression for Miller-effect Capacitance reflected at the input of CE amplifier.

(b) What are main criteria for a circuit to oscillate? Discuss comprehensively about phase shift oscillator and find out its frequency of oscillation.

Q.3 (a) Illustrate that with input 2 grounded, signals at input 1 in the ideal Differential Amplifier leads to in phase signal at output 2 by finding out dc voltages and currents in the circuit.

(b) For inverting configuration of Op-Amp, find out the expression for close loop gain (V_o/V_i)?

Short Questions

Max Marks: 20

Q.1 Draw frequency response curve of a voltage amplifier, with properly labeled regions. Why series and shunt capacitors are ineffective in the mid frequency region?

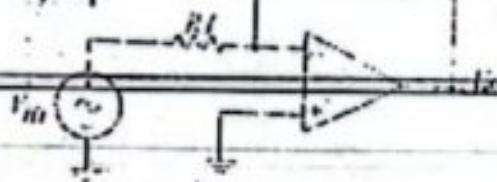
Q.2 Signify the role of zener knee current and zener maximum current from the reverse biased operating characteristics of the Zener diode.

Q.3 Calculate the efficiency of a class B amplifier for a supply voltage, of $V_{CC} = 24\text{ V}$ with peak reverse voltage of $V_{L(p)} = 22\text{ V}$.Q.4 Calculate the ripple voltage of a capacitor filtered power supply. Filter is of $100\text{ }\mu\text{F}$ value connected to a load drawing a current of 50 mA .

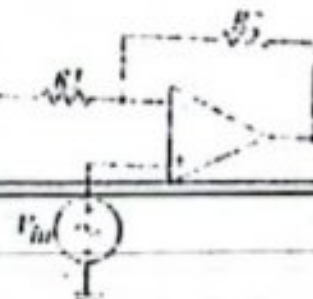
Q.5 Discuss the basic principle of a photodiode.

Q.6 For the following circuits, find out the gain when $R_1 = 1\text{ K}\Omega$ and $R_2 = R_F = 1\text{ M}\Omega$.

$$A = -\frac{R_2}{R_1} = -\frac{1\text{ M}\Omega}{1\text{ K}\Omega} = -1000$$



(a) Inverting amplifier



(b) Non-inverting amplifier

$$A = 1 + \frac{R_f}{R_1} = 1 + \frac{1\text{ M}\Omega}{1\text{ K}\Omega}$$

$$A = 1001$$